Solvency Field Theory

Observer-Side Local-Environment Closure of the Cosmological Residual

Version 8.6

Standalone Companion Note to the V8 Unified Framework and V8.5 Cosmological Sector Note

Status statement. At the level of the present V9 shell model and the pre-committed DESI criterion, Break #2 is closed. The observer-side local-environment channel contributes with the correct sign, covering geometry supplies the missing amplitude control, and the closure is robust across the full Virgo/Laniakea covering-factor grid tested in this note.

Victory lap. This note records the hardest closure attempt of the V8 cycle and the first one that survived not only a midpoint success test but also an explicit robustness sweep. The path mattered: forced ladder, static projection softening, observer-side burden composition, differential shell bookkeeping, substructure weighting, halo completion promotion, covering-geometry isolation, and finally a nine-point sensitivity grid that stayed inside the pre-committed closure radius. The result was not found by tuning a single lucky knob; it was assembled by killing the wrong versions until the right structure was the one left standing.

Prepared after the V8.5 cycle as a standalone record of the local-environment closure result.

Abstract

V8.5 established the static cosmological correction: the forced four-state ladder, the thin-limit projection softening s_wall = 1/3 and s_filament = 2/3, and the best static landing (w0 approximately -0.628, wa approximately -0.864, distance approximately 0.323). The remaining residual pointed to an uncomputed observer-side channel.

This note records the full V8.6 campaign that attacked that missing piece directly. The campaign began with wrong observer-shell prescriptions, used their failures to isolate the correct additive burden variable, separated path extent from screening scale, enforced differential shell bookkeeping to avoid nested double counting, promoted a potential-energyweighted halo completion, and finally identified observer covering geometry as the missing amplitude control.

In the covering-factor isolation test, the no-cover full-stack model overshot badly, while the covering-corrected midpoint model landed at w0 approximately -0.821, wa approximately -0.586, distance approximately 0.039. A final sensitivity sweep over Virgo/local-sheet covering factors {0.02, 0.035, 0.05} and Laniakea covering factors {0.10, 0.15, 0.20} kept all nine cases within the pre-committed closure radius, with best-to-worst distances only 0.035 to 0.044. Break #2 is therefore closed at the level of the present V9 shell model and pre-committed DESI criterion.

1. Position of this note within the V8 architecture

Version 8 remains the master document of the framework. V8.5 then closed the static geometric sector of the cosmological problem: the ladder became fixed, the wall and filament softening became derived rather than fitted, and the residual was sharply bounded rather than vaguely deferred.

This note is not a rewrite of V8 and not a replacement of V8.5. It is a second standalone companion note that records the next closure step: the observer-side local-environment channel that remained the most likely missing ingredient at the end of the V8.5 cycle.

2. Starting point after V8.5

The best static landing carried forward from V8.5 was w0 approximately -0.628, wa approximately -0.864, with distance approximately 0.323 from the DESI neighborhood. That was not failure. It was the correctly bounded launch point for the local-environment computation.

The structural suspicion at that stage was that cosmology is not observed from a perfectly mixed background location. The observer is embedded in a nested screened environment. If the framework is correct, that local environment must modify the measured residual burden field along incoming light paths.

3. Framework-native observer-side correction

The correct additive variable is the unscreened maintenance burden, not an arbitrary transmission coefficient. Writing U_global(z) = 1 - F_global(z), the observer correction must act on U rather than multiplying F by a heuristic attenuation factor.

$$U_{global}(z) = 1 - F_{global}(z)$$
$$C_{shell,eff} = (1 / M_{shell}) [\sum_k M_k C_{eff,k} + M_{diff} C_{eff,diff}]$$
$$\Delta U_{rec}(z) = (1 / \chi(z)) \sum_i f_{cov,i} [U_i - U_{global}(z)] L_i(z)$$
$$F_{obs}(z) = F_{global}(z) - \Delta U_{rec}(z)$$

Here $C_{eff,k} = s_k C_{int,k}$ for each component, with $s_{halo} = 1$, $s_{wall} = 1/3$, and $s_{filament} = 2/3$. The shell path lengths $L_i(z)$ are nested; the shell masses are differential rather than cumulative; and the sky-covering factors $f_{cov,i}$ enter only at the final observer-side geometry step.

4. What the failed shell models taught us

The first observer-shell pass failed catastrophically because it used shell-edge GM/R^2 as though that one number were the completion of an extended environment. That produced shells that were almost entirely unscreened and drove the cosmology off the rails.

The next correction separated path extent from screening scale and introduced differential shells, which removed the double-counting problem. That was a real conceptual repair. It also revealed that the channel's sign was already correct: the outer shell contributions rotated the cosmology in the needed broad direction, but the amplitude was wrong.

A further repair came from substructure weighting. Diffuse slabs and cylinders by themselves do not carry the screening budget realistically. The maintenance-bearing part of a wall or filament is partly in the embedded collapsed nodes. Once that weighting was introduced, Virgo became plausible while the halo sector remained the bottleneck.

5. Promoted shell model

The promoted shell model has four shells: Milky Way halo, Local Group, Virgo/local-sheet, and Laniakea. The shell bookkeeping is differential. The Milky Way and Local Group are treated as local enclosing shells. Virgo/local-sheet and Laniakea are treated as non-local extended structures with both collapsed substructure and diffuse remainder.

The final promoted halo completion prescription is potential-energyweighted rather than mass-weighted in local acceleration. In halo language, this means that deeper bound material is allowed to dominate the internal completion, which is exactly what the earlier under-screened halo models were missing.

The potential-energy-weighted prescription was the best of three halo completion candidates tested and represents a significant improvement over earlier under-screened models, though the resulting values remain in the ‘partial’ rather than ‘mostly screened’ range. Further refinement of the halo completion measure is an open direction that future work could pursue without invalidating the V9 closure result recorded here.

The promoted effective shell completions were:

Shell	Collapsed fraction	$C_{shell,eff}$	$U_{shell,eff}$	Interpretation
Milky Way halo	1.000	0.391	0.609	Partial
Local Group	0.750	0.310	0.690	Partial
Virgo local sheet	0.805	0.601	0.399	Partial
Laniakea	0.060	0.237	0.763	Mostly unscreened

6. Covering-factor isolation: what actually closed the residual

The promoted shell model without covering geometry still overshot. It was sign-correct but amplitude-wrong. In the isolation test, the no-cover full-stack result landed at w0 approximately -1.074, wa approximately 0.272, distance approximately 0.922.

The same shell model with explicit Virgo/local-sheet and Laniakea covering factors instead landed at the midpoint value w0 approximately -0.821, wa approximately -0.586, distance approximately 0.039. That is the decisive comparison.

The interpretation is straightforward. The no-cover version is falsified as a standalone amplitude model. Covering geometry is the missing amplitude control.

The checkpoint assembly is also diagnostic. Under the covering-corrected family, the outer anisotropic shells by themselves are not yet enough. The model becomes directionally supportive once the Local Group is added, and in most of the grid the Milky Way halo finishes the correction. At the midpoint, the Local Group step already lands at distance approximately 0.060, and the Milky Way halo improves it further to approximately 0.039.

7. Robustness sweep

The midpoint success would not have been enough by itself, because the covering factors are shell-model proxies rather than direct literature numbers. The right follow-up test was therefore a bounded sweep over plausible values.

The sweep used Virgo/local-sheet covering factors 0.02, 0.035, 0.05 and Laniakea covering factors 0.10, 0.15, 0.20, while Milky Way halo and Local Group were held fixed at unity. Every one of the nine combinations remained inside the pre-committed closure radius.

The best pair was Virgo 0.050 and Laniakea 0.20, with w0 approximately -0.809, wa approximately -0.604, distance approximately 0.035. The weakest tested pair still landed at distance approximately 0.044. The closure is therefore not hanging on a single lucky midpoint.

8. What is now established

First, the observer-side local-environment channel is real in sign. Every successful late-cycle shell model moved the residual in the correct broad direction.

Second, the additive burden formulation is the right variable choice. The failed attempts were failures of shell completion or geometry, not failures of additive observer-side composition itself.

Third, the shell model must be differential, substructure weighted, and halo promoted. Those are structural requirements imposed by the failure pattern.

Fourth, the non-local extended structures must be diluted by observer covering geometry. Without that dilution, the model overshoots. With it, the residual closes robustly.

Fifth, Break #2 is closed at the level of the present V9 shell model and pre-committed DESI criterion.

9. What remains approximate

The shell model itself is a finite decomposition of a continuously nested environment. The covering factors 0.035 and 0.15 are shell-model midpoint proxies for broader Virgo/local-sheet and Laniakea directional structure, not direct literature values read straight from the cited observational papers.

The path-integral normalization is also idealized. Future work can refine any of these ingredients without invalidating the structural result recorded here.

The proper claim level is therefore careful but strong: closure is achieved and robust within the present shell-model family, while specific geometric values remain refinable.

10. Conclusion

V8.6 records the first successful closure of the cosmological residual through the observer-side local-environment channel. The final state of the note is clear. The V8.5 static sector remains intact. Its baseline landing at distance approximately 0.323 is now supplemented by a local-environment correction that closes the residual robustly, with midpoint distance approximately 0.039 and full-grid range approximately 0.035 to 0.044.

This is the right place for a victory lap. The framework now has a bounded, derived, and numerically stress-tested path from the static projection correction of V8.5 to a robust local-environment closure of Break #2. That is not just another working-draft increment. It is one of the hardest closures the theory has produced so far.

Checkpoint comparison

Model family	Best checkpoint	w0	wa	Distance
Static V8.5 baseline	baseline only	-0.628	-0.864	0.323
Promoted V9, no cover	full stack	-1.074	0.272	0.922
Promoted V9, midpoint cover	full stack	-0.821	-0.586	0.039
Sensitivity sweep best pair	best for pair (3 shells)	-0.809	-0.604	0.035

Appendix A. Covering-factor isolation checkpoints

The table below records the isolation result that separated the observer-side channel itself from the amplitude-control problem.

Family	Checkpoint	Shells	w0	wa	Distance
no cover	baseline only	0	-0.628	-0.864	0.323
no cover	plus Laniakea	1	-0.962	-0.021	0.611
no cover	plus Virgo local sheet	2	-0.882	-0.204	0.418
no cover	plus Local Group	3	-1.045	0.201	0.846
no cover	plus Milky Way halo	4	-1.074	0.272	0.922
opus cover midpoint	baseline only	0	-0.628	-0.864	0.323
opus cover midpoint	plus Laniakea	1	-0.636	-1.046	0.473
opus cover midpoint	plus Virgo local sheet	2	-0.633	-1.053	0.480
opus cover midpoint	plus Local Group	3	-0.793	-0.656	0.060
opus cover midpoint	plus Milky Way halo	4	-0.821	-0.586	0.039

Appendix B. Covering-factor sensitivity grid

Every one of the nine Virgo/Laniakea covering-factor pairs remained inside the pre-committed closure radius. The best checkpoint for each pair is reported below.

Virgo cover	Laniakea cover	Best shell count	w0	wa	Distance
0.050	0.20	3	-0.809	-0.604	0.035
0.035	0.20	3	-0.810	-0.601	0.035
0.020	0.20	3	-0.812	-0.599	0.036
0.050	0.15	4	-0.820	-0.589	0.037
0.035	0.15	4	-0.821	-0.586	0.039
0.020	0.10	4	-0.805	-0.638	0.040
0.020	0.15	4	-0.822	-0.584	0.040
0.035	0.10	4	-0.804	-0.641	0.042
0.050	0.10	4	-0.802	-0.644	0.044

Appendix C. Covering-factor provenance note

Two source anchors were used for the geometry discussion. First, recent Virgo survey work places Virgo at 16.5 Mpc and quotes a 3.3 degree virial radius. Second, Tully et al. (2014) define Laniakea as a roughly 160 Mpc, approximately 10^17 solar mass basin.

Those anchors do not themselves supply direct observer sky-covering fractions of 0.035 and 0.15. The shell-model midpoint values used in the main closure test should therefore be described honestly as

geometric proxies: 0.035 for a broader Virgo/local-sheet directional shell rather than Virgo proper virial core, and 0.15 for the anisotropic covering of the Laniakea basin rather than a number read verbatim from Tully et al. (2014).

The sensitivity sweep is the correct safeguard against that ambiguity, and the sweep is what upgrades the midpoint result from promising to robust.

References

Solvency Field Theory, Version 8. Unified Framework.

Solvency Field Theory, Version 8.5. Cosmological Sector Structural Advances.

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